

# ANGELA ZHOU

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## POSITIONS

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|------------------------------------------------------------------------|--------------------|
| <b>Assistant Professor, USC Marshall Data Science &amp; Operations</b> | <i>2022-</i>       |
| Department of Computer Science, by courtesy                            | <i>2025-</i>       |
| Affiliated Faculty, USC Center for AI and Society                      | <i>2023-</i>       |
| <b>Research Fellow at Simons Institute Program on Causality</b>        | <i>Spring 2022</i> |
| <b>Postdoc at Foundations of Data Science Institute, UC Berkeley</b>   | <i>2021- 2022</i>  |
| <b>PhD Student at Cornell University and Cornell Tech</b>              |                    |
| <b>Department of Operations Research and Information Engineering</b>   | <i>2016-2021</i>   |

## RESEARCH INTERESTS

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My research develops principled methodology for responsible data-driven decision-making in public-interest operations. I unify tools from causal inference, operations management, optimization, and statistical machine learning to study how algorithmic systems shape decisions, access, and outcomes in social services and other high-stakes settings. My use-inspired basic research is motivated by partnership and co-design with non-profits and public agencies, through which I identify, formalize, and solve fundamental challenges that arise in practice.

## EDUCATION

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|------------------------------------------------------------------------------|------------------|
| <b>Princeton University</b>                                                  | <i>2012-2016</i> |
| Undergraduate: Class of 2016, Operations Research and Financial Engineering. |                  |

## PROFESSIONAL EXPERIENCE

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|                                    |      |
|------------------------------------|------|
| Microsoft Research New York City   | 2019 |
| PlaceIQ Data Science               | 2016 |
| AppNexus Data Science/Optimization | 2015 |

*Note:* Authorship order follows field conventions. The default order is equal-contribution alphabetical order, denoted with “*With...*”. Some projects involving interdisciplinary authors are in contributinal order, where all authors are listed and my main or first-authorship is underlined. A superscript <sup>†</sup> indicates a student co-author at the time of collaboration. Additionally selective designations (e.g., Oral, featured certification) are **highlighted in dark blue**.

## ACCEPTED JOURNAL PUBLICATIONS

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|                                                                        |                  |
|------------------------------------------------------------------------|------------------|
| <b>Reduced-Rank Outcome Compression for Causal Policy Optimization</b> | <u>TMLR</u> 2026 |
| <i>With E. Nwankwo<sup>†</sup>.</i>                                    |                  |
| Accepted with minor revision.                                          |                  |
| <b>Multi-Accurate CATE is Robust to Unknown Covariate Shifts</b>       | <u>TMLR</u> 2024 |
| <b>(<b>featured certification</b>)</b> .                               |                  |
| <i>With C. Kern and M. Kim</i>                                         |                  |

**An Empirical Evaluation of the Impact of New York’s Bail Reform on Crime Using Synthetic Controls** Statistics and Public Policy 2023.

*A. Zhou, A. Koo, N. Kallus, R. Ropac, R. Peterson, S. Koppel, T. Bergin.*

**Assessing Algorithmic Fairness with Unobserved Protected Class Using Data Combination** Management Science 2021.

*With N. Kallus and X. Mao*

**Minimax-Optimal Policy Learning Under Unobserved Confounding** Management Science 2021.

*With N. Kallus*

Supersedes earlier conference version NeurIPS at 2018, “Confounding-Robust Policy Improvement”.

**UNDER REVIEW**

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**Robust Fitted-Q-Evaluation and Iteration under Sequentially Exogenous Unobserved Confounders**

*With D. Bruns-Smith<sup>†</sup>.*

Major Revision at Management Science.

**Data-Driven Influence Functions for Optimization-Based Causal Inference**

*With M. I. Jordan and Y. Wang.*

Under second-round review at Journal of Machine Learning Research after “Rejection with invitation to resubmit.” Re-submitted 03/2026. Supersedes conference version *Empirical Gateaux Derivatives* below.

**Mind the Gap: Optimal and Equitable Encouragement Policies**

*Solo-authored paper*

**WORKING PAPERS**

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**Optimizing and Learning Assortment Decisions in the Presence of Platform Disengagement**

*With M. Sumida.*

**Batch-Adaptive Causal Annotations**

*With L. Goldkind, E. Nwankwo<sup>†</sup>*

Conference version accepted at AISTATS 2026; journal version to supersede conference version.

**Structured Difference-of-Q via Orthogonal Learning**

*With D. Cao<sup>†</sup>*

Conference version accepted at AISTATS 2026; journal version to supersede conference version.

**Bridging Predictions and Interventions in Social Systems: A Primer.**

*L. Liu\*, D. Raji<sup>†</sup>\*, A. Zhou\*, et al.*

## Due Process on Hold: A Queueing Framework for Improving Access in SNAP

With A. Daw and C. Pache<sup>†</sup>

## Bridging Predictions and Interventions in Social Systems

L. Liu\*, D. Raji<sup>†</sup>\*, A. Zhou\*, et al.

## IN PROGRESS

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### Engagement in Freemium Games

With S. Demir<sup>†</sup>. and M. Sumida.

### Batch-Adaptive Surrogate Validation

With L. Goldkind, E. Nwankwo<sup>†</sup>

## REFEREED CONFERENCE PUBLICATIONS

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The primary publishing venues for machine learning are selective “top-tier” refereed conferences (e.g. [NeurIPS](#) (Conference on Neural Information Processing Systems, 20–25% acceptance rates), [ICML](#) (International Conference on Machine Learning, 20–25%), [AISTATS](#) (International Conference on Artificial Intelligence and Statistics, 25–30%), [FAccT](#) (ACM Conference on Fairness, Accountability, and Transparency; formerly known as FAT\*, 25%)).

### Batch-Adaptive Causal Annotations

[AISTATS](#) 2026

With L. Goldkind, E. Nwankwo<sup>†</sup>

### Structured Difference-of-Q via Orthogonal Learning

[AISTATS](#) 2026

With D. Cao<sup>†</sup>

### Fostering the Ecosystem of AI for Social Impact Requires Expanding and Strengthening Evaluation Standards [NeurIPS](#) 2025 Position Paper Track (6% acceptance rate).

With B. Wilder.

### Reward-Relevance-Filtered Linear Offline Reinforcement Learning [AISTATS](#) 2024.

Solo-authored paper.

### Optimal and Fair Encouragement Policy Evaluation and Learning [NeurIPS](#) 2023.

Solo-authored paper

### Empirical Gateaux Derivatives for Causal Inference [NeurIPS](#) 2022, [Oral-designated](#).

Journal version submitted.

With M. Jordan and Y. Wang.

### Off-Policy Evaluation with Policy-Dependent Optimization Response [NeurIPS](#) 2022.

With W. Guo<sup>†</sup> and M. Jordan.

## Stateful Offline Contextual Policy Evaluation and Learning

AISTATS 2022

*With N. Kallus*

## It's COMPASlicated: The Messy Relationship between RAI Datasets and Algorithmic Fairness Benchmarks

NeurIPS 2021 Datasets and Benchmarks Track, [Oral](#)

*M. Bao<sup>†</sup>\*, [A. Zhou](#)\*, S. Zottola\*, B. Brubach, S. Desmarais, A. Horowitz, K. Lum, S. Venkatasubramanian*

Contributions statement: MB, AZ, SZ are equal-contribution first-authors and wrote the paper. BB, SD, AH, KL, SV are equal-contribution advising authors.

## Fairness, Welfare, and Equity in Personalized Pricing

FAccT 2021.

*With N. Kallus*

## Confounding-Robust Policy Evaluation in Infinite-Horizon Reinforcement Learning

*With N. Kallus.*

NeurIPS 2020

## Assessing Disparate Impacts of Personalized Interventions: Identifiability and Bounds

*With N. Kallus*

NeurIPS 2019.

## The Fairness of Risk Scores Beyond Classification: Bipartite Ranking and the xAUC Metric

*With N. Kallus*

NeurIPS 2019.

## Interval Estimation of Individual-Level Causal Effects

AISTATS 2019.

*With N. Kallus and X. Mao*

## Residual Unfairness in Fair Machine Learning from Prejudiced Data

ICML 2018

*With N. Kallus*

## Policy Evaluation and Optimization with Continuous Treatments

AISTATS 2018

*With N. Kallus*

## TEACHING

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### DSO 670: Bridging Predictions and Interventions in Social Systems

Spring 2025

I developed a PhD seminar course drawing on our workshop and whitepaper series of the same name. The first half of the course taught the fundamentals of causal inference and decision-making and responsible AI. The second half of the course facilitated student presentations and projects of select topics on reliable causal data-driven decision-making.

### BUAD 311: Business Analytics

Fall 2022-2025

I teach the undergraduate core operations management class. Each section has 70 students. I taught 4 sections in Fall 2022, 5 sections in Fall 2023 and Fall 2024, and 2 sections in Fall 2025.

## HONORS/AWARDS

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|                                                                                                                        |      |
|------------------------------------------------------------------------------------------------------------------------|------|
| Rising Star in AI (Harvard Center for Research on Computation and Society)                                             | 2021 |
| Rising Star in Data Science (University of Chicago Center for Data and Computing)                                      | 2020 |
| Winner, INFORMS Data Mining Best Paper Award (Confounding-Robust Policy Improvement)                                   | 2018 |
| Finalist for Best Paper of INFORMS Data Mining and Decision Analytics Workshop (Confounding-Robust Policy Improvement) | 2017 |
| National Defense Science and Engineering Graduate Fellowship                                                           | 2016 |
| Ahmet S. Cakmak Thesis prize winner for undergraduate thesis                                                           | 2016 |

## MENTORSHIP

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### Advising PhD students

|              |                      |
|--------------|----------------------|
| Woojin Chae  | USC (2025 - present) |
| Luyang Zhang | USC (2022 - 2024)    |

### Co-authorship with PhD students

|                                                                                            |                            |
|--------------------------------------------------------------------------------------------|----------------------------|
| David Bruns-Smith                                                                          | UC Berkeley (2022-present) |
| <i>First placement:</i> Tenure-track Assistant Professor at MIT Sloan Finance and MIT EECS |                            |
| Defu Cao                                                                                   | USC (2022-present)         |
| <i>First placement:</i> Tenure-track Assistant Professor at Baylor University              |                            |
| Sebnem Demir                                                                               | USC (2024-present)         |
| James Enouen                                                                               | USC (2025-present)         |
| Ezinne Nwankwo                                                                             | UC Berkeley (2023-present) |
| Wenshuo Guo                                                                                | UC Berkeley (2022)         |

### Undergraduate research mentorship

|                                                                                                 |                                |
|-------------------------------------------------------------------------------------------------|--------------------------------|
| Xiao Qi Lee                                                                                     | via JumpStart (2025)           |
| <i>Project:</i> Auditing LLMs for Algorithmic Fairness in Casenote-Augmented Tabular Prediction |                                |
| Chloe Pache (co-advised)                                                                        | via DSO Summer Scholars (2025) |
| <i>Project:</i> Due Process on Hold: A Queueing Framework for Improving Access in SNAP          |                                |
| Michael Qin                                                                                     | via DSO Summer Scholars (2025) |
| <i>Project:</i> Housing Placement Transition Prediction                                         |                                |
| <i>First placement:</i> UW ISOM PhD program                                                     |                                |
| Jonas Liang (co-advised)                                                                        | via DSO Summer Scholars (2025) |
| <i>Project:</i> Can We Predict What Didn't Happen? Counterfactuals from Panel Data              |                                |
| Yongjia Wang (co-advised)                                                                       | via DSO Summer Scholars (2024) |
| <i>Project:</i> "The Fragility of Gerrymandering Metrics"                                       |                                |

Mengchu Yue (co-advised) via DSO Summer Scholars (2024, 2026 - present)  
*Project*: “The Fragility of Gerrymandering Metrics”

Joshua Zhong (co-advised) via DSO Summer Scholars (2024)  
*Project*: “The Fragility of Gerrymandering Metrics”

### **Masters Students**

Yash Patel, on Local LLM serialized classification USC (2025)  
Aryan Parab, on LLM summary hallucination detection USC (2026)

## **GRANTS/FUNDING AWARDS**

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USC iORB (Institute for Outlier Research in Business) 2024–2025  
“Human-Centered Operations: Developing AI to Combine Qualitative User Experience and Quantitative Data for Better Decisions.” \$10,000  
Continuing funding in 2025 \$5,600

Microsoft Accelerate Foundation Models Research Program 2023  
Awarded \$20,000 in Azure credits through a competitive call for research proposals on research related to foundation models.

## **SERVICE WITHIN USC**

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SoCal OR/OM day co-organizer; contributed \$5,000 funding via iORB conference grant 2026  
DSO Summer Scholars Co-organizer 2025  
OCCS Teaching Grants Committee 2025

### **Student committees**

#### Thesis proposal / defense committees

UC Berkeley EECS (outside member): Ezinne Nwankwo (2025)

Computer science: Sid Devic (proposal & defense, 2025–26), James Enouen (proposal, 2025), Defu Cao (proposal, 2025)

DSO: Sebnem Demir (proposal, 2026)

ISE: Bill Tang (dissertation defense, 2025)

Economics: Wesley Huang (dissertation defense, 2025), Zhan Gao (dissertation defense, 2024)

#### Qualifying exam committees

DSO: Sebnem Demir (2024), Tianmin Xie (2024), Yanfei Zhou (2023)

Computer Science: Defu Cao (2024)

Economics: Paul Delatte (2024), Zhan Gao (2024), Wesley Huang (2023)

## **INVITED TALKS AND PRESENTATIONS**

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(\*) indicates presentation by a junior coauthor.

**Batch-Adaptive Annotations for Causal Inference with Complex-Embedded Outcomes**

|                                                               |         |
|---------------------------------------------------------------|---------|
| Econometric Society Interdisciplinary Frontiers: AI/ML (*)    | 06/2026 |
| INI Causal Inference workshop, Causality and machine learning | 06/2026 |
| Center for AI and Society, USC                                | 02/2026 |
| University of Pennsylvania, Wharton OID                       | 04/2025 |
| USC ShowCAIS                                                  | 04/2025 |

**Structure in Offline Reinforcement Learning: Difference-of-Q functions**

|                                                                      |         |
|----------------------------------------------------------------------|---------|
| INI Causal Inference workshop, Identification and Discovery          | 03/2026 |
| INFORMS Applied Probability Society                                  | 07/2025 |
| CIFAR Workshop: World Models: Causality, Neuroscience, and AI Safety | 06/2024 |

**Robust Fitted-Q-Iteration under Unobserved Confounders**

|                                                                |         |
|----------------------------------------------------------------|---------|
| USC ISE Seminar                                                | 12/2025 |
| Online Causal Inference Seminar                                | 05/2025 |
| UCLA Statistics Seminar                                        | 02/2024 |
| INFORMS Annual Meeting (*)                                     | 10/2023 |
| USC Machine Learning Lunch                                     | 10/2023 |
| Machine Learning and Friends Seminar, UMass Amherst            | 09/2023 |
| Harvard/MIT Econometrics Department Seminar                    | 09/2023 |
| Bravo Center/JEA/SNSF Workshop on Using Data to Make Decisions | 07/2023 |

**Optimal and Fair Encouragement Policy Evaluation and Learning**

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|---------------------------------------------------------------------------|---------|
| INFORMS Manufacturing and Service Operations Management Conference (MSOM) | 07/2025 |
| Econometric Society Interdisciplinary Frontiers: AI/ML                    | 06/2024 |
| Berkeley N-of-1 Workshop                                                  | 07/2024 |
| American Causal Inference Conference (poster)                             | 05/2024 |
| Cal Poly San Luis Obispo Economics Seminar                                | 04/2024 |
| NeurIPS (poster)                                                          | 11/2023 |
| Mechanism Design for Social Good, Inequality Working Group                | 10/2023 |
| ShowCAIS, University of Southern California                               | 04/2023 |

**Optimizing and Learning Assortment Decisions in the Presence of Platform Disengagement**

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|---------------------------------------------------------------------------|---------|
| INFORMS Annual Meeting                                                    | 10/2023 |
| INFORMS Manufacturing and Service Operations Management Conference (MSOM) | 07/2023 |

**Empirical Gateaux Derivatives for Causal Inference and Stochastic Optimization**

|                                                                          |         |
|--------------------------------------------------------------------------|---------|
| Econometric Society Interdisciplinary Frontiers: AI/ML                   | 2024    |
| American Causal Inference Conference ( <a href="#">oral</a> )            | 2024    |
| International Conference on Stochastic Programming                       | 08/2023 |
| Stanford University, Operations and Information Technology               | 2023    |
| Southern California OR/OM Day                                            | 2023    |
| Simons Institute, Causality Reunion                                      | 2023    |
| USC Econometrics Reading Group                                           | 2023    |
| INFORMS Annual Meeting                                                   | 2022    |
| Adobe Research                                                           | 2022    |
| Keynote, CPAIOR masterclass on Machine Learning and Optimization         | 2022    |
| Joint Statistical Meetings (on confounding-robust infinite-horizon RL)   | 2022    |
| American Causal Inference Conference (on Stateful Off-Policy Evaluation) | 2022    |

**Minimax-Optimal Policy Learning under Unobserved Confounding:**

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|--------------------------------------------------------------------------|------|
| University of California, Irvine, Statistics                             | 2023 |
| University of Michigan, Computer Science and Engineering                 | 2023 |
| Engelhardt Lab, Gladstone                                                | 2022 |
| Simons Workshop on Algorithmic Advances in Causal Inference              | 2022 |
| Simons Fellow Seminar Series                                             | 2022 |
| Berkeley Biostatistics Seminar                                           | 2022 |
| NYU Langone                                                              | 2022 |
| INFORMS Annual Meeting                                                   | 2021 |
| Berkeley Causal Inference Research Group; BLISS; Semi-Autonomous Systems | 2021 |
| INFORMS Healthcare conference                                            | 2021 |
| Center for Causal Inference Seminar                                      | 2021 |
| ANU HMI Seminar Series                                                   | 2021 |
| Health Data Science Workshop                                             | 2021 |
| Harvard CRCS AI for Social Impact                                        | 2021 |
| Northwestern IEMS                                                        | 2021 |
| USC Marshall School of Business, Operations                              | 2021 |



|                                                             |      |
|-------------------------------------------------------------|------|
| UNC Kenan-Flagler School of Business                        | 2021 |
| Cornell Johnson School of Business                          | 2021 |
| Stanford Management Science and Engineering                 | 2021 |
| MIT Sloan/Schwarzman                                        | 2021 |
| UBC Sauder Operations and Logistics                         | 2020 |
| Berkeley Haas (Operations and IT)                           | 2020 |
| Columbia IEOR                                               | 2020 |
| University of Minnesota ISYE                                | 2020 |
| Columbia Biostatistics Causal Inference Learning Group      | 2020 |
| Facebook Core Data Science                                  | 2020 |
| Kellogg-Wharton OM Workshop                                 | 2020 |
| Duke Fuqua Workshop on Operations Research and Data Science | 2019 |

**Confounding-Robust Policy Evaluation in Infinite-Horizon Reinforcement Learning:**

INFORMS 2020.

**Assessing Algorithmic Unfairness with Unobserved Protected Class:**

|                                                    |      |
|----------------------------------------------------|------|
| HMI DAIS Seminar at Australian National University | 2021 |
| Experian DataLab Brazil                            | 2020 |
| CMU Fairness/Ethics/Accountability Reading Group   | 2020 |

**Assessing Fairness of Personalized Interventions:**

|         |      |
|---------|------|
| INFORMS | 2019 |
|---------|------|

**Confounding-Robust Policy Improvement:**

|                                  |      |
|----------------------------------|------|
| INFORMS Conference on Healthcare | 2019 |
| Princeton                        | 2019 |
| MSR NYC                          | 2018 |
| INFORMS                          | 2018 |

**Residual Unfairness:**

|                                          |      |
|------------------------------------------|------|
| Crime Lab New York (UChicago Urban Labs) | 2018 |
|------------------------------------------|------|

**Policy Evaluation and Optimization with Continuous Treatments:**

|         |      |
|---------|------|
| Spotify | 2017 |
| INFORMS | 2017 |

## EXTERNAL SERVICE AND REFEREEING

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### **Journal refereeing:**

Management Science (MS)

Operations Research (OR)

Manufacturing & Service Operations Management (M&SOM)

Marketing Science

Journal of Machine Learning Research (JMLR)

Annals of Statistics (AOS)

Journal of the American Statistical Association (JASA)

Journal of the Royal Statistical Society: Series B (JRSS:B)

Biometrika

Annals of Applied Statistics (AOAS)

Journal of Causal Inference

Journal of Business and Economic Statistics (JBES)

Quantitative Economics

INFORMS Journal on Computing

Naval Research Logistics

Statistics in Medicine

### **Program co-chair:**

ACM Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO) (2022).

ACM EAAMO is a new conference that aims to highlight work where techniques from algorithms, optimization, and mechanism design, along with insights from the social sciences and humanistic studies, can help improve equity and access to opportunity for historically disadvantaged and underserved communities.

The role of the program co-chair was to oversee the paper submission and reviewing process (130 reviewers for 160 papers), convene and delegate to conference sub-chair team to support programming, and curate the academic program of keynotes and sessions.

### **(Funded) Major Workshop Co-organizing**

Bridging Prediction and Intervention Problems in Social Systems Banff, 2024

Bridging Prediction and Intervention Problems in Social Systems Simons Institute, 2026

A 1-week workshop with \$35k budget from Simons and substantial additional funding from the MacArthur Foundation.

### **Senior Program Committee:**

Theory Area Chair for EAAMO 2021, Area Chair for EAAMO 2025-26

Area chair for FAccT 2023-26

Area chair for AISTATS 2024-25

Area chair for UAI 2026

**Awards for refereeing:**

Management Science Meritorious Service Award 2023

Top reviewer at NeurIPS, ICML, AISTATS (top 400, top 5%, 33%, 10%).

**Conference review:**

NeurIPS 18-24 (and Datasets and Benchmarks 21)

ICML 18-24

AISTATS 19/22

EC 23

FAccT 20/22-23

MD4SG '20

Conference on Causal Learning and Reasoning (CLeaR) 22

Workshops: NeurIPS 2021 Workshops, Theoretical Foundations of Reinforcement Learning (ICML 2020), Workshop on Reinforcement Learning Theory (ICML 2021), Causal Inference for Sequential Decision-Making (NeurIPS 2021), Strategic Machine Learning (NeurIPS 2021)

**Senior Conference Organization:**

UAI Scientific Integrity Chair 2023

Workflow co-chair, American Causal Inference Conference 2022

Tutorials chair for FAccT 2023

**Other:**

NSF Panelist (2021, 2024, 2026)

Judge, INFORMS Applied Probability Society Student Paper Competition (2021-22, 25)

Judge, INFORMS Public Sector Operations Research Student Paper Competition (2023)